

Today

Open Methods: Newton-Raphson
+ Friends

Secant, Brent's, Fixed Point Iter
(maybe)

Reminders

Lab 1 due tomorrow

First: Errors (Chapra's notation)

E : absolute error (units)

ϵ : relative (percent) error

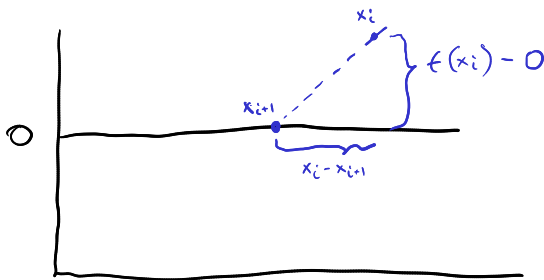
Subscripts

t : true (typically unknown)

a : approximate

s : stopping

Newton-Raphson: use $f'(x)$



$$f'(x_i) = \frac{f(x_i) - 0}{x_i - x_{i+1}} \quad \text{rise over run}$$

$$(x_i - x_{i+1}) f'(x_i) = f(x_i)$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Newton-Raphson

input: function f and its derivative f' , initial guess x_0 , tolerance ϵ_s , maximum iterations N

output: estimate x_r of the root

$x_r \leftarrow x_0$

for $i \leftarrow 1$ to N do

$x_{r,old} \leftarrow x_r$

$x_r \leftarrow x_{r,old} - \frac{f(x_{r,old})}{f'(x_{r,old})}$ ← update

$\epsilon_a \leftarrow \left| \frac{x_r - x_{r,old}}{x_r} \right| \cdot 100\%$

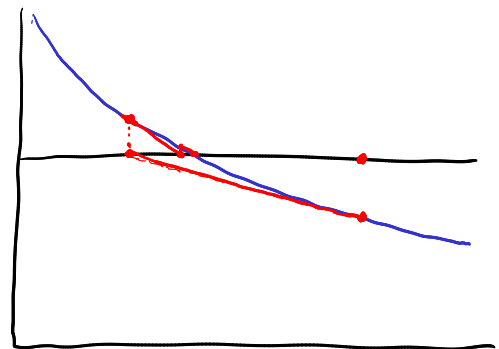
if $\epsilon_a < \epsilon_s$ then

 return x_r

end

end

return x_r // did not converge in N iterations



Beauty of Newton-Raphson: Extremely fast convergence near root!

Bisection

$$E_{a,i+1} = \frac{1}{2} E_{a,i}$$

"linear convergence"

number of sig figs increases linearly with i

(different terminology from $E_{a,i} = O((\frac{1}{2})^k)$)

Guaranteed to converge if

 - Δx contains root

 - $f(x)$ is continuous

fast

Newton Raphson

Near Root

$$\rightarrow E_{t,i+1} \approx \frac{-f''(x_r)}{2f'(x_r)} E_{t,i}^2 \quad \text{"can be shown"}$$

"quadratic convergence"

number of sigfigs doubles every iteration

Can diverge

FAST near root

Downside: Newton requires $f'(x)$

Secant Method

$$f'(x_i) \approx \frac{f(x_{i-1}) - f(x_i)}{x_{i-1} - x_i}$$

plug into Newton update

$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$

Secant Method

input : function f , initial guesses $\underline{x_{-1}}$ and $\underline{x_0}$, tolerance ε_s ,
maximum iterations N

output: estimate x_r of the root

$$x_{\text{prev}} \leftarrow x_{-1}; \quad x_r \leftarrow x_0$$

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for  $i \leftarrow 1$  to  $N$  do

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$$x_{r,\text{old}} \leftarrow x_r$$

$$x_r \leftarrow x_{r,\text{old}} - \frac{f(x_{r,\text{old}}) (x_{\text{prev}} - x_{r,\text{old}})}{f(x_{\text{prev}}) - f(x_{r,\text{old}})}$$

$$x_{\text{prev}} \leftarrow x_{r,\text{old}}$$

$$\varepsilon_a \leftarrow \left| \frac{x_r - x_{r,\text{old}}}{x_r} \right| \cdot 100\%$$

if $\varepsilon_a < \varepsilon_s$ then

| return x_T

end

end

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return  $x_T$            // did not converge in  $N$  iterations
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